

# Adenovirus-36 Is Associated with Obesity in Children and Adults in Sweden as Determined by Rapid ELISA

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Background Experimental and natural human adenovirus-36 (Adv36) infection of multiple animal species results in obesity through increasing adipogenesis and lipid accumulation in adipocytes. Presence of Adv36 antibodies detected by serum neutralization assay has previously been associated with obesity in children and adults living in the USA, South Korea and Italy, whereas no association with adult obesity was detected in Belgium/the Netherlands nor among USA military personnel. Adv36 infection has also been shown to reduce blood lipid levels, increase glucose uptake by adipose tissue and skeletal muscle biopsies, and to associate with improved glycemic control in non-diabetic individuals. Principal Findings Using a novel ELISA, 1946 clinically well-characterized individuals including 424 children and 1522 non-diabetic adults, and 89 anonymous blood donors, residing in central Sweden representing the population in Stockholm area, were studied for the presence of antibodies against Adv36 in serum. The prevalence of Adv36 positivity in lean individuals increased from ~7% in 1992–1998 to 15–20% in 2002–2009, which paralleled the increase in obesity prevalence. We found that Adv36-positive serology was associated with pediatric obesity and with severe obesity in females compared to lean and overweight/mildly obese individuals, with a 1.5 to 2-fold Adv36 positivity increase in cases. Moreover, Adv36 positivity was less common among females and males on antilipid pharmacological treatment or with high blood triglyceride level. Insulin sensitivity, measured as lower HOMA-IR, showed a higher point estimate in Adv36-positive obese females and males, although it was not statistically significant ( $p = 0.08$ ). Conclusion Using a novel ELISA we show that Adv36 infection is associated with pediatric obesity, severe obesity in adult females and lower risk of high blood lipid levels in non-diabetic Swedish individuals.